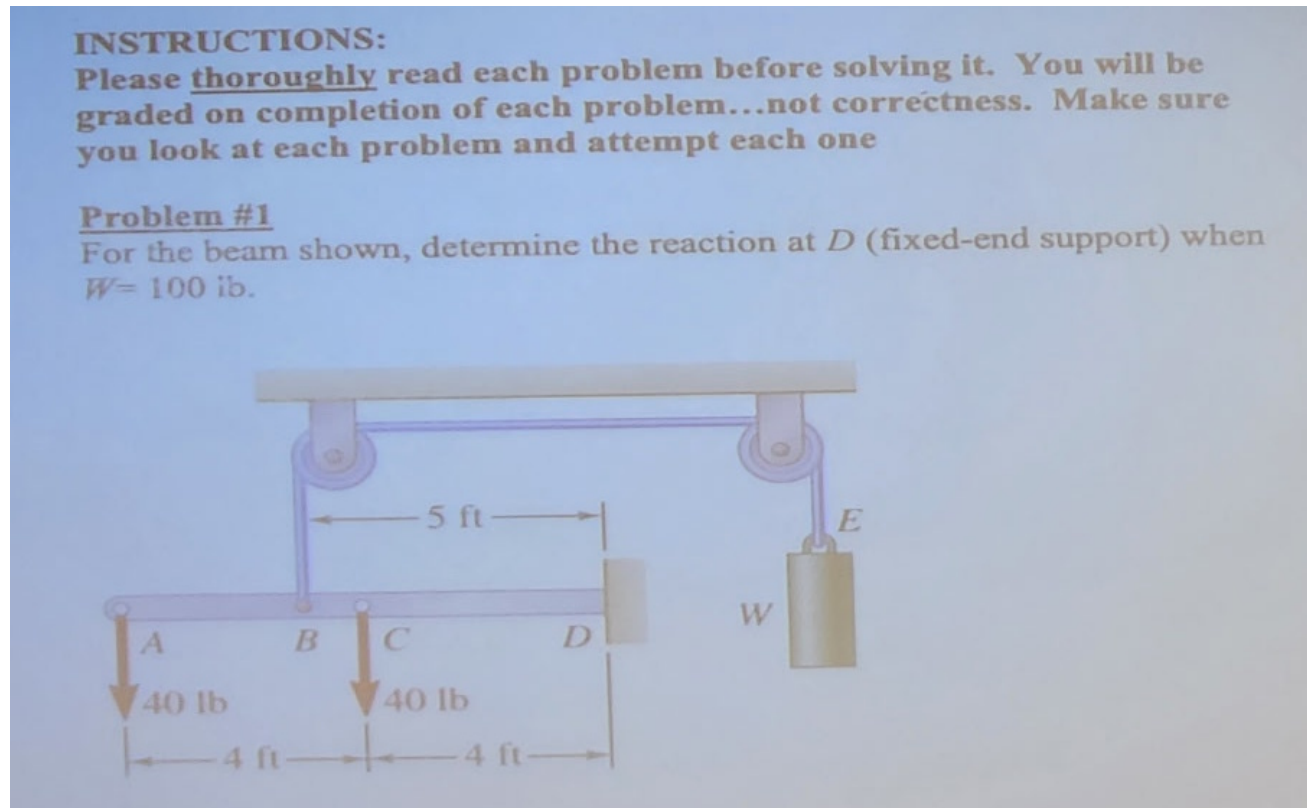


CE 333 Statics - Test 2 (A) - ANSWER KEY

Problem #1 (10 points)

For the beam shown, determine the reaction at D (fixed-end support) when $W = 100$ lb.



Answer

$$V_D = -20 \text{ lb (downward)}, M_D = -120 \text{ lb} \cdot \text{ft (clockwise)}, H_D = 0$$

Solution

Given: Beam A-C-D with total length 8 ft. Loads: 40 lb at A, 40 lb at C. Pulley at B (midway between A and C) applies $W = 100$ lb upward. Fixed support at D.

Vertical force equilibrium:

$$\sum F_y = 0 : V_D + 100 - 40 - 40 = 0$$

$$V_D = -20 \text{ lb}$$

Negative indicates downward reaction.

Horizontal force equilibrium:

$$\sum F_x = 0 : H_D = 0$$

Moment equilibrium about D:

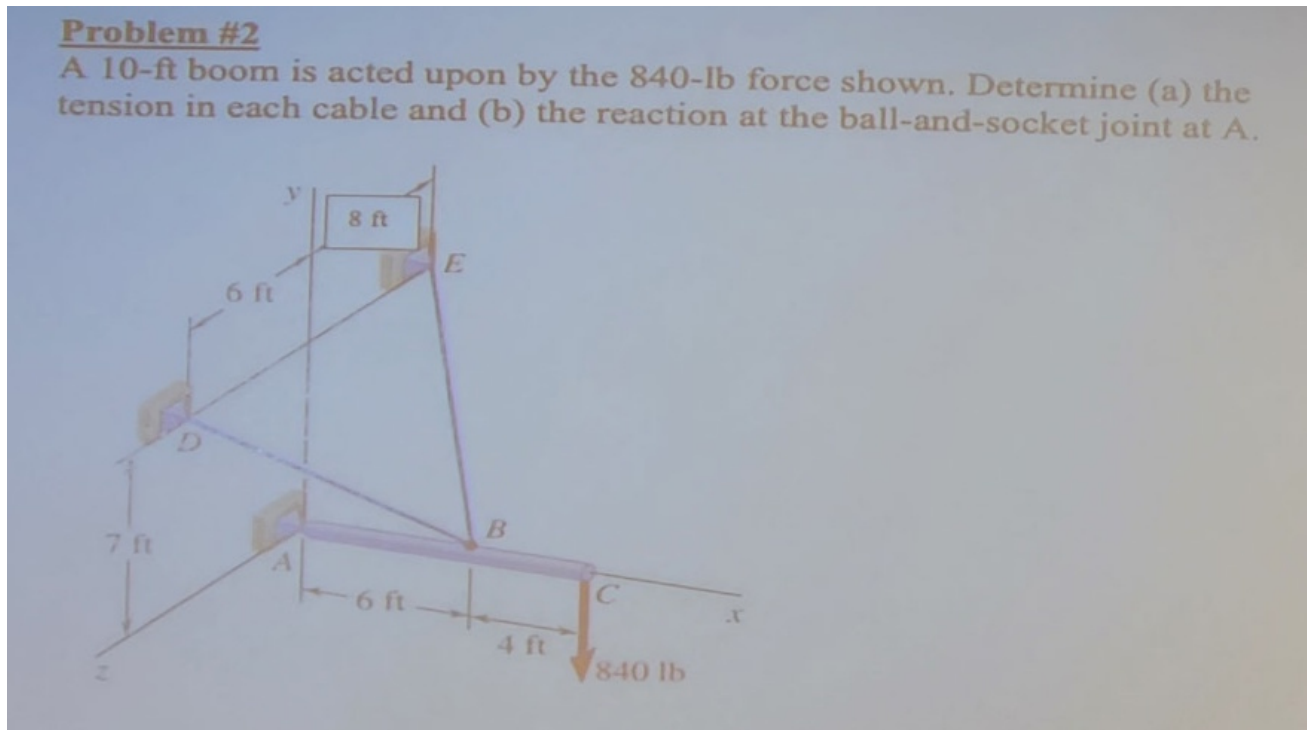
$$\sum M_D = 0 : M_D + 100(6) - 40(8) - 40(4) = 0$$

$$M_D = -600 + 320 + 160 = -120 \text{ lb} \cdot \text{ft}$$

Problem #2 (15 points)

A 10-ft boom is acted upon by the 840 lb force shown. Determine

- (a) the tension in each cable and
- (b) the reaction at the ball-and-socket joint at A.



Answer

(a) $T_{BD} = 1,257 \text{ lb}$, $T_{BE} = 1,046 \text{ lb}$; (b) $\vec{R}_A = (1,200\hat{i} - 560\hat{j}) \text{ lb}$

Solution

Coordinate system: x = horizontal, y = vertical, z = in/out of screen

Positions: A at $(0, 0, 0)$, B at $(6, 0, 0)$, C at $(10, 0, 0)$, D at $(0, 7, -6)$, E at $(0, 7, 8)$

Load: 840 lb downward at C = $(0, -840, 0) \text{ lb}$

Step 1: Unit vectors for cables

Cable BD:

$$\vec{r}_{BD} = D - B = (-6, 7, -6)$$

$$|\vec{r}_{BD}| = 11 \text{ ft}$$

$$\hat{u}_{BD} = \frac{1}{11}(-6, 7, -6)$$

Cable BE:

$$\vec{r}_{BE} = E - B = (-6, 7, 8)$$

$$|\vec{r}_{BE}| = \sqrt{149} \approx 12.21 \text{ ft}$$

$$\hat{u}_{BE} = \frac{1}{\sqrt{149}}(-6, 7, 8)$$

Step 2: Moment equilibrium about A

$$\sum \vec{M}_A = \vec{0}$$

The y-component and z-component equations:

$$-\frac{36T_{BD}}{11} + \frac{48T_{BE}}{\sqrt{149}} = 0$$

$$\frac{42T_{BD}}{11} + \frac{42T_{BE}}{\sqrt{149}} - 8,400 = 0$$

Step 3: Solve for tensions

From first equation: $T_{BD} = 1.202T_{BE}$

Substituting: $T_{BE} = 1,046 \text{ lb}$, $T_{BD} = 1,257 \text{ lb}$

Problem #3 (12 points)

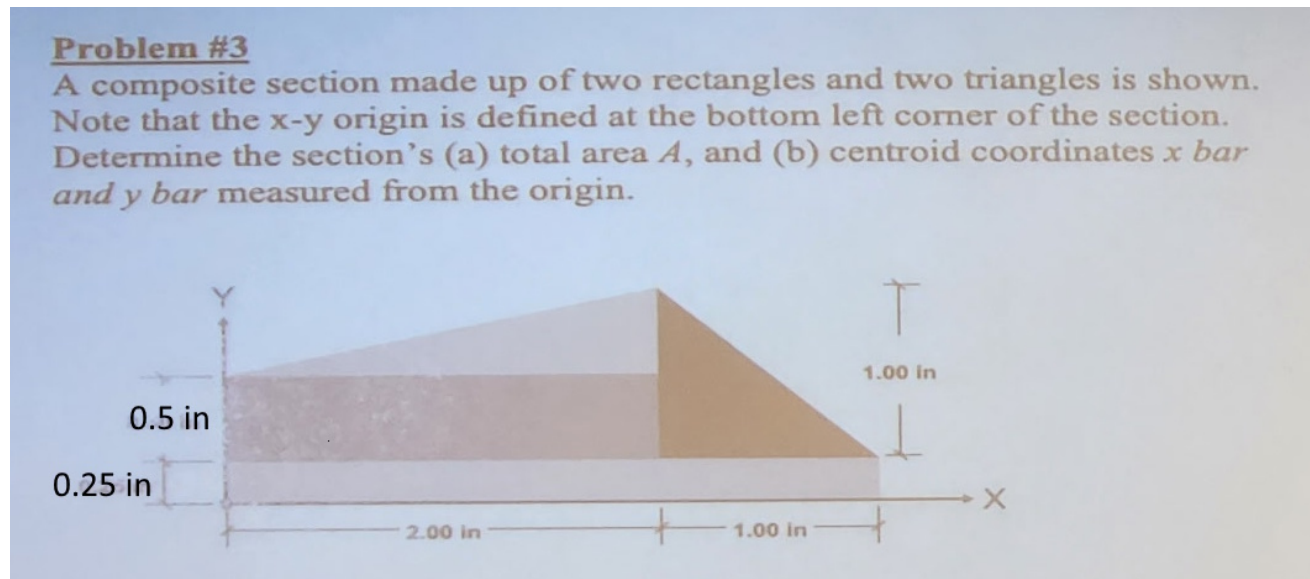
A composite section made up of two rectangles and two triangles is shown.

The x-y origin is defined at the bottom-left corner of the section.

Determine the section's

(a) total area A , and

(b) centroid coordinates $\bar{x}$ and $\bar{y}$ measured from the origin.



Answer

(a) $A = 2.25 \text{ in}^2$; (b) $\bar{x} = 1.54 \text{ in}$, $\bar{y} = 0.40 \text{ in}$

Solution

Dimensions from diagram:

- Rectangle 1 (bottom): Width = 3.00 in, Height = 0.25 in
- Rectangle 2 (middle): Width = 2.00 in, Height = 0.25 in
- Triangle 3 (top left): Base = 2.00 in, Height = 0.5 in
- Triangle 4 (top right): Base = 1.00 in, Height = 1.00 in

Individual areas:

$$A_1 = (3.00)(0.25) = 0.75 \text{ in}^2$$

$$A_2 = (2.00)(0.25) = 0.50 \text{ in}^2$$

$$A_3 = \frac{1}{2}(2.00)(0.5) = 0.50 \text{ in}^2$$

$$A_4 = \frac{1}{2}(1.00)(1.00) = 0.50 \text{ in}^2$$

Total area:

$$A = 0.75 + 0.50 + 0.50 + 0.50 = 2.25 \text{ in}^2$$

Centroid calculation:

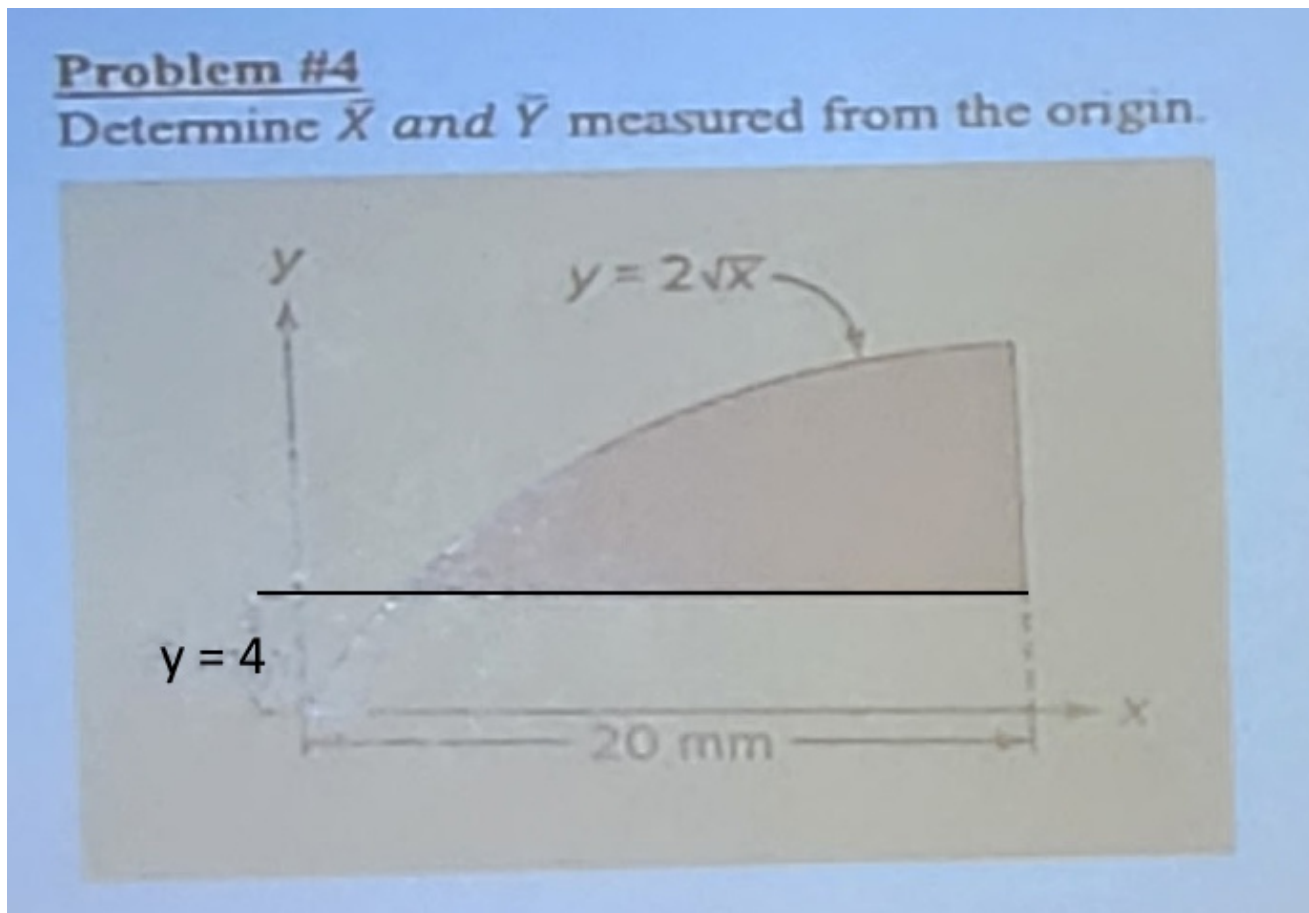
Shape	A_i	x_i	y_i	$A_i x_i$	$A_i y_i$
1	0.75	1.50	0.125	1.125	0.094
2	0.50	1.00	0.375	0.500	0.188
3	0.50	1.33	0.667	0.667	0.333
4	0.50	2.33	0.583	1.167	0.292
Σ	2.25			3.459	0.907

$$\bar{x} = \frac{3.459}{2.25} = 1.54 \text{ in}$$

$$\bar{y} = \frac{0.907}{2.25} = 0.40 \text{ in}$$

Problem #4 (15 points)

Determine $\bar{X}$ and $\bar{Y}$ measured from the origin for the area bounded by $y = 2\sqrt{x}$ and $y = 4$, with x ranging from 0 to 20 mm.



Answer

$$\bar{X} = 1.20 \text{ mm}, \bar{Y} = 3.0 \text{ mm}$$

Solution

Find intersection point:

$$2\sqrt{x} = 4 \Rightarrow \sqrt{x} = 2 \Rightarrow x = 4 \text{ mm}$$

Calculate area:

$$A = \int_0^4 (4 - 2\sqrt{x}) dx$$

$$A = \left[4x - \frac{4}{3}x^{3/2} \right]_0^4$$

$$A = 16 - \frac{4}{3}(8) = \frac{16}{3} \text{ mm}^2$$

Calculate $\bar{X}$:

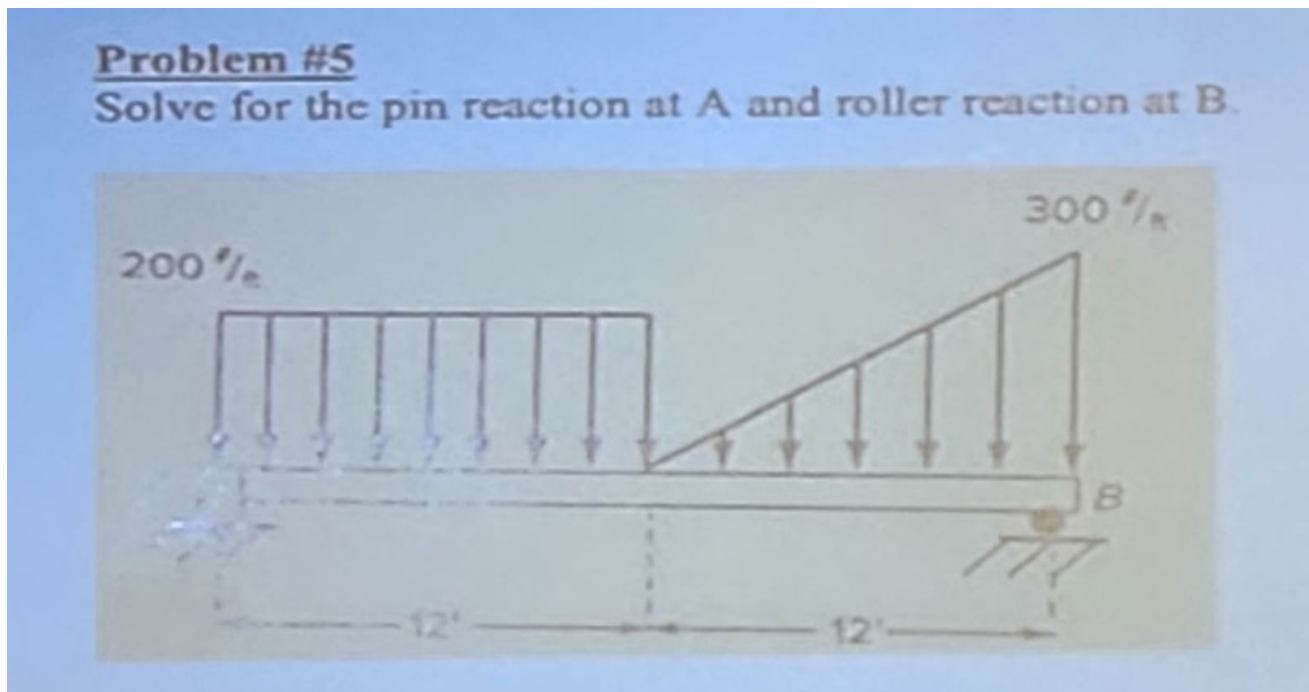
$$\begin{aligned}\bar{X} &= \frac{1}{A} \int_0^4 x(4 - 2\sqrt{x}) \, dx \\&= \frac{3}{16} \int_0^4 (4x - 2x^{3/2}) \, dx \\&= \frac{3}{16} \left[2x^2 - \frac{4}{5}x^{5/2} \right]_0^4 \\&= \frac{3}{16} \left[32 - \frac{128}{5} \right] = \frac{6}{5} = 1.20 \text{ mm}\end{aligned}$$

Calculate $\bar{Y}$:

$$\begin{aligned}\bar{Y} &= \frac{1}{A} \int_0^4 \frac{1}{2} [(4)^2 - (2\sqrt{x})^2] \, dx \\&= \frac{3}{32} \int_0^4 (16 - 4x) \, dx \\&= \frac{3}{32} [16x - 2x^2]_0^4 = \frac{3}{32}(32) = 3.0 \text{ mm}\end{aligned}$$

Problem #5 (13 points)

Solve for the pin reaction at A and the roller reaction at B for the beam subjected to the distributed loads shown.



Answer

$$R_A = 2,100 \text{ lb}, R_B = 2,100 \text{ lb}$$

Solution

Convert distributed loads to equivalent point loads:

Uniform load (0 to 12 ft):

- $W_1 = 200 \times 12 = 2,400 \text{ lb}$ at $x = 6 \text{ ft}$ from A

Triangular load (12 to 24 ft):

- $W_2 = \frac{1}{2}(12)(300) = 1,800 \text{ lb}$
- Centroid of triangle: $\frac{2}{3}$ from zero end = 8 ft from left edge
- From A: $x_2 = 12 + 8 = 20 \text{ ft}$

Sum moments about A:

$$\sum M_A = 0 : R_B(24) = 2,400(6) + 1,800(20) = 14,400 + 36,000 = 50,400$$

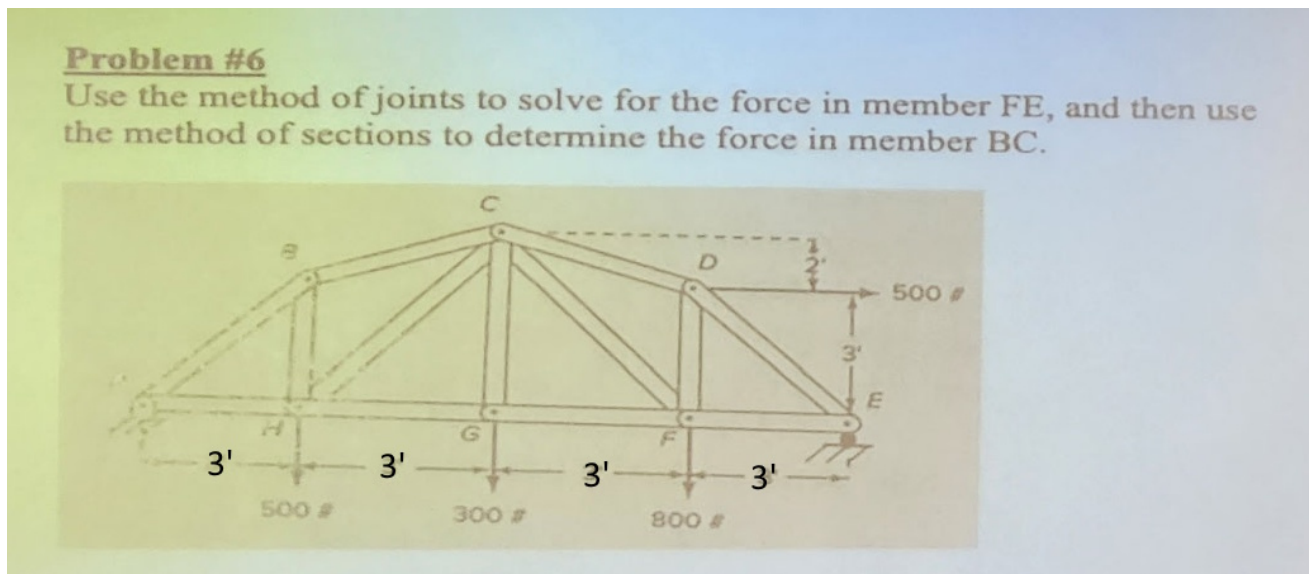
$$R_B = 2,100 \text{ lb}$$

Sum vertical forces:

$$\sum F_y = 0 : R_A = 4,200 - 2,100 = 2,100 \text{ lb}$$

Problem #6 (15 points)

Use the method of joints to solve for the force in member FE, and then use the method of sections to determine the force in member BC.



Answer

$$F_{FE} = 875 \text{ lb (compression); } F_{BC} = 672 \text{ lb (compression)}$$

Solution

Step 1: Support reactions

$$\sum F_y = 0 : A_y + B_y = 500 + 300 + 800 = 1,600 \text{ lb}$$

$$\sum M_A = 0 : B_y(12) - 500(3) - 300(6) - 800(9) = 0$$

$$B_y = \frac{10,500}{12} = 875 \text{ lb}$$

$$A_y = 1,600 - 875 = 725 \text{ lb}$$

Step 2: Force in member FE (method of joints)

At joint E: unknowns are F_{FE} (horizontal) and F_{DE} (diagonal)

Geometry: Member DE has rise = 3 ft, run = 3 ft, so $\theta = 45^\circ$

Vertical equilibrium:

$$\sum F_y = 0 : 875 + F_{DE} \sin(45^\circ) = 0$$

$$F_{DE} = -\frac{875}{\sin(45^\circ)} = -1,237 \text{ lb (compression)}$$

Horizontal equilibrium:

$$\sum F_x = 0 : F_{FE} - F_{DE} \cos(45^\circ) = 0$$

$$F_{FE} = -1,237 \times 0.707 = -875 \text{ lb (compression)}$$

Step 3: Force in member BC (method of sections)

Cut through BC, CG, BG. Take moments about G at (6, 0).

Distance from G to line BC: $d = 3\sqrt{2} = 4.243 \text{ ft}$

Moment equilibrium:

$$\sum M_G = 0 : A_y(6) - 500(3) - F_{BC}(4.243) = 0$$

$$725(6) - 1,500 = F_{BC}(4.243)$$

$$F_{BC} = \frac{2,850}{4.243} = 672 \text{ lb (compression)}$$
